

# NAVRAJ SINGH

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## EDUCATION & PROFESSIONAL TRAINING

**Indian Institute of Science Education and Research (IISER) Mohali**

Punjab, India

*BS-MS Dual Degree Student (Fourth Year)*

Oct 2022 – Present

CPI: 8.9/10

**Note:** Detailed reports and code for the projects available at [Github](#).

## AI SAFETY & ALIGNMENT RESEARCH

**Evaluation Awareness in Large Language Models**

*Pre-print on arXiv*

*Lead Researcher | Supervisor: Maheep Chaudhary*

Feb – June 2026

- Mapped the representational scaling of evaluation awareness across compact architectures (Gemma 3, Phi-3, Llama-3 8B). Utilized Chain-of-Thought analysis, representation probing, and Integrated Gradients attribution for the investigation.
- Engineered a dual-pathway mitigation pipeline combining automated prompt sanitization with activation counter-steering, to neutralize evaluation artifacts across 200 high-confidence prompts, achieving an average behavioral flip rate of 70.58% across all evaluated architectures.

**Interpretability Frameworks for Diffusion Models**

*Ongoing MS Thesis*

*Research Student | Supervisor: Prof. Ashutosh Modi*

May 2026 – Present

- Building mechanistic evaluation tools to analyze feature representation and concept formulation during the reverse denoising process.
- Applying the Schrödinger bridging framework to track multi-step trajectories from Gaussian noise to structured outputs, identifying precise layers where early generation choices stabilize.

**Narrow Secret Loyalty Extension Using CAI**

*Research Fellow | Bluedot Technical AIS Project | Funded by Bluedot Impact*

June - July 2026

- Selected for a 1-month research track focusing on the mechanistic evaluation and vulnerability probing of proprietary/specialized Loyalty models to detect hidden alignment failures.

**Bluedot Technical AI Safety Course**

*Scholar – 6-Day Intensive Cohort*

June 2026

- Selected for a technical training cohort covering alignment theory, mechanistic interpretability foundations, and frontier model vulnerability profiles.

**VEIL: Mechanistic Upstream Guardrails for Generative Biology**

*Apart Research Hackathon*

*Core Developer | AIXBio Track*

- Trained an  $L_1$ -regularized linear probe on ProteinMPNN decoder latents to identify structural markers of pathogenic risk, isolating causal features independent of token length.
- Analyzed feature polysemanticity and exposed generalization boundaries on unseen biological targets (e.g., Ricin) where safety vectors conflated structural threat with structural rigidity.

**Mechanistic Analysis of Simplicity Bias and Feature Steering**

*PRECOC Lab*

*Independent Researcher*

- Trained Sparse Autoencoders (SAEs) on penultimate activations to disentangle compressed neural features.

- Proved a mechanistic variance between simple and robust features via targeted feature steering interventions, applying token-level gradients to successfully manipulate downstream classifications.

## Replication Studies

- Completed replication studies of various foundation papers in the field of Foundational DL and AI Safety.

## OTHER TECHNICAL & QUANTITATIVE INTERNSHIPS

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### Fully Homomorphic Encryption in AI (CKKS)

*Supervisor: Prof. Mridula Verma*

Jul 2025 – Nov 2025

- Implemented encrypted neural networks running evaluations directly on ciphertexts to ensure strict data privacy.
- Managed CKKS noise budgets and scaling factors across deep architectures to optimize the precision-compute tradeoff.

### Valid Circuit Detection in Hexa-board PCB

*Supervisor: Prof. Satyajit Jena*

May 2025 – Jul 2025

- Engineered an end-to-end computer vision object-detection pipeline leveraging a hybrid SSD architecture (ResNet-50 backbone paired with an auxiliary ResNet-18 classifier) for multi-scale anomaly detection.

### Computational Methods for Option Pricing

*Supervisor: Prof. Anindya Goswami*

Jun 2024 – Aug 2024

- Developed high-performance C++ implementations of continuous-time option pricing and stochastic algorithms.

### Grain Size Distribution Analysis Using Mask-RCNN

*Supervisor: Prof. Yunus Ali Pulpadan*

Jun 2024 – Aug 2024

- Fine-tuned a custom Mask-RCNN framework on drone telemetry to segment and calculate granular distribution frequencies.

## TECHNICAL SKILLS

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- **Languages & Libraries:** Python, C++, PyTorch, TensorFlow, CUDA, R, NumPy, Scikit-learn, Pydantic, aiosqlite.
- **Core Competencies:** Mechanistic Interpretability, Sparse Autoencoders (SAEs), Circuit Discovery, Linear Probing & Steering, Causal Interventions (Activation Patching), Fully Homomorphic Encryption (CKKS).
- **Mathematics:** Linear Algebra, Measure Theoretic Probability, Cryptography & Number Theory, Group Theory, Stochastic PDEs.

## ACHIEVEMENTS & RELEVANT COURSEWORK

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- **Achievements:** JEE Main (Top 1 Percentile), IISER Aptitude Test (AIR-1345), KVPY Written Test (Qualified), SOF Science Olympiad (State Rank 51), SOF Mathematics Olympiad (State Rank 102).
- **Key Courses:** Machine Learning (Stanford CS229), Deep Learning, Quantum Information & Computation, Theory of Computation, Measure Theoretic Probability, Functional Analysis, Non-Linear Dynamics and Chaos.